

TOLEDO EXECUTIVE, OH, USA

WMO: 724287

Lat: 41.563N

Lon: 83.476W

Elev: 190

StdP: 99.07

Time Zone: -5.00 (NAE)

Period: 98-19

WBAN: 04848

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-17.1	-14.1	-21.6	0.6	-16.3	-18.4	0.8	-13.2	12.5	-2.2	11.6	-2.1	3.7	230	0.530

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.2	32.9	23.3	31.6	22.6	30.0	21.9	25.0	30.6	24.1	28.9	23.3	27.9	5.3	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
23.2	18.4	27.9	22.5	17.6	27.0	22.0	17.0	26.3	77.0	30.5	73.3	28.9	69.9	27.8	27.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 11.2	9.7	8.6	-20.3	35.1	3.2	2.0	-22.6	36.6	-24.5	37.8	-26.3	38.9	-28.7	40.4		
(5)			-20.4	26.3	3.1	0.8	-22.6	26.9	-24.4	27.3	-26.2	27.8	-28.4	28.4		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	10.8	-3.3	-1.9	3.2	9.9	16.4	21.7	23.7	22.6	19.0	12.2	5.8	-0.4	(6)
(7)		DBStd	10.71	6.35	6.08	5.91	4.93	4.75	3.67	3.08	2.72	4.01	4.84	5.19	5.72	(7)
(8)		HDD10.0	1540	412	335	225	61	4	0	0	0	1	30	147	325	(8)
(9)		HDD18.3	3269	669	566	470	257	96	12	1	2	39	199	377	581	(9)
(10)		CDD10.0	1832	1	2	14	57	203	349	425	389	270	99	19	3	(10)
(11)		CDD18.3	521	0	0	1	3	36	112	168	134	58	9	0	0	(11)
(12)		CDH23.3	4421	0	0	6	47	343	957	1517	1014	461	74	1	0	(12)
(13)		CDH26.7	1479	0	0	1	7	105	343	571	293	144	16	0	0	(13)
(14)	Wind	WSAvg	4.0	4.8	4.6	4.4	4.7	4.0	3.6	3.2	2.9	3.1	3.9	4.3	4.5	(14)
(15)	Precipitation	PrecAvg	855	52	52	65	81	83	92	82	80	73	58	69	67	(15)
(16)		PrecMax	1214	117	146	145	161	173	215	233	215	206	159	182	173	(16)
(17)		PrecMin	560	7	7	16	22	24	7	9	10	15	7	14	14	(17)
(18)		PrecStd	133	27	33	30	33	38	42	43	46	41	32	37	32	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	14.7	18.1	23.4	27.7	32.0	34.3	35.3	33.0	32.8	28.8	22.4	17.0	(19)
(20)			MCWB	12.5	13.2	17.1	17.1	21.5	23.2	24.5	23.7	23.4	20.8	15.2	14.5	(20)
(21)		2%	DB	11.3	13.0	18.7	24.3	29.5	32.4	33.3	31.3	30.5	25.3	18.1	13.2	(21)
(22)			MCWB	9.6	10.9	13.3	15.8	20.4	22.5	23.6	23.3	21.9	18.2	13.7	11.4	(22)
(23)		5%	DB	8.0	9.1	15.8	21.6	27.4	30.6	31.9	29.7	28.0	22.6	16.3	10.2	(23)
(24)			MCWB	6.3	6.4	11.8	14.6	19.5	21.8	23.1	22.3	20.6	17.1	12.5	8.1	(24)
(25)		10%	DB	4.9	5.9	12.6	18.6	24.7	28.5	29.9	28.1	26.0	20.2	13.8	7.2	(25)
(26)			MCWB	3.2	3.6	9.3	13.1	18.1	20.6	22.1	21.5	19.6	15.7	10.6	5.4	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	12.8	14.5	17.7	19.1	23.1	25.1	26.3	25.7	24.8	21.9	16.8	14.9	(27)
(28)			MCD B	14.4	17.1	23.0	24.8	29.5	32.1	32.8	31.0	30.9	27.7	19.7	16.9	(28)
(29)		2%	WB	9.9	10.7	14.6	17.2	21.7	23.8	25.3	24.8	23.3	19.9	14.7	11.7	(29)
(30)			MCD B	11.1	12.7	17.9	22.4	27.4	29.9	31.3	29.2	27.9	23.4	17.6	13.0	(30)
(31)		5%	WB	6.6	6.7	12.3	15.4	20.4	22.9	24.3	23.7	22.1	18.0	12.8	8.1	(31)
(32)			MCD B	8.1	8.7	15.2	20.6	25.6	28.6	29.1	27.6	26.4	21.5	15.8	9.7	(32)
(33)		10%	WB	3.3	3.6	9.5	13.6	19.1	21.9	23.4	22.8	20.9	16.2	10.8	5.5	(33)
(34)			MCD B	4.7	5.7	12.6	18.1	23.7	26.8	28.0	26.6	24.4	19.7	13.7	7.1	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	7.4	7.9	9.2	10.7	10.6	10.2	9.8	10.6	9.9	8.5	6.8	(35)	
(36)			MCD BR	9.9	12.1	13.9	15.1	13.4	12.8	11.9	11.5	13.2	13.0	12.0	9.6	(36)
(37)			MCWBR	8.7	9.7	9.6	8.6	6.7	5.5	4.8	5.2	6.1	7.2	8.8	8.3	(37)
(38)		5% WB	MCD BR	9.6	11.5	13.0	13.6	11.9	11.1	10.3	9.8	10.6	11.2	10.8	9.2	(38)
(39)			MCWBR	8.8	9.8	9.8	8.9	6.6	5.6	4.7	5.1	5.6	6.8	8.7	8.5	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.302	0.335	0.359	0.404	0.435	0.451	0.444	0.436	0.388	0.362	0.325	0.312	(40)	
(41)		taud	2.409	2.299	2.320	2.217	2.181	2.184	2.222	2.229	2.342	2.395	2.457	2.407	(41)	
(42)		Ebn at Noon	848	865	886	865	845	829	831	826	845	826	814	800	(42)	
(43)		Edh at Noon	80	104	114	136	145	145	138	134	111	94	76	74	(43)	
(44)	All-Sky Solar Radiation	RadAvg	1.53	2.44	3.48	4.63	5.44	6.06	6.07	5.33	4.34	2.87	1.83	1.26	(44)	
(45)		RadStd	0.16	0.32	0.31	0.39	0.47	0.41	0.34	0.20	0.34	0.23	0.25	0.12	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(47) Regional (70 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+95	N/A

Nomenclature: See separate page